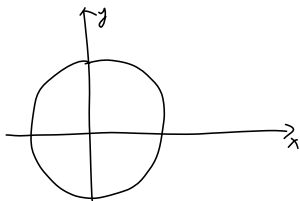


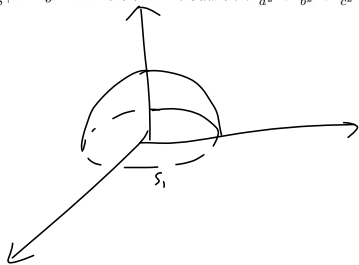


1. 设 $L: x^2 + y^2 = 1$, 求 $\int_L (x+y)^2 dl$.



$$\begin{aligned} \begin{cases} x = \cos \theta \\ y = \sin \theta \end{cases} & \int_0^{2\pi} (\sin \theta + \cos \theta)^2 \sqrt{\sin^2 \theta + \cos^2 \theta} d\theta \\ &= \int_0^{2\pi} 1 + \sin 2\theta d\theta \\ &= \theta - \frac{1}{2} \cos 2\theta \Big|_0^{2\pi} = 2\pi \end{aligned}$$

2. 计算 $\int_{S^+} x^3 dy \wedge dz$, 其中 S^+ 为椭圆面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 的上半部分, 正方向向下, $a, b, c > 0$.



$$\begin{aligned} & \int_{S^+ \cup S_1} p dy_1 dz + q dz_1 dx + r dx_1 dy \\ \stackrel{\text{高斯}}{=} & \iiint_V \left(\frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} + \frac{\partial r}{\partial z} \right) dx dy dz \quad \frac{\partial(x, y, z)}{\partial(\rho, \varphi, \theta)} = abc \rho^2 \sin \varphi d\rho d\varphi d\theta \\ &= \iiint_V 3x^2 dx dy dz \quad \begin{cases} \frac{x}{a} = \rho \cos \varphi \\ \frac{y}{b} = \rho \sin \varphi \cos \theta \\ \frac{z}{c} = \rho \sin \varphi \sin \theta \end{cases} \\ &= abc \iiint 3\rho^2 \cos^2 \varphi \rho^2 \sin \varphi d\rho d\varphi d\theta \\ &= abc \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \cos^2 \varphi \sin \varphi d\varphi \int_0^1 \rho^4 d\rho \\ &= abc \cdot 2\pi \cdot \left[-\frac{\cos^3 \varphi}{3} \right]_0^{\frac{\pi}{2}} \cdot \frac{\rho^5}{5} \Big|_0^1 \\ &= 2\pi abc \cdot \int_0^1 t^2 dt \cdot \frac{1}{5} = \frac{2\pi abc}{5} \cdot \frac{t^3}{3} \Big|_0^1 = \frac{2\pi abc}{15} \end{aligned}$$

$$\int_{S_1} x^3 dy \wedge dz = 0$$

$$\therefore \int_{S_1} x^3 dy \wedge dz = \frac{2\pi abc}{15}$$

3. 设 $D = \{(x, y) : x > 0\}$

(1) 若 $A, B \in D$, $L(A, B)$ 为 D 内连接 A, B 两点的逐段光滑曲线, 问 $\int_{L(A, B)} \frac{ydx - xdy}{x^2 + 2y^2}$ 是否与路径 L 的选取无关?

(2) 是否存在 $\mathbb{R}^2$ 上的二元函数 $z = z(x, y)$, 使得 $dz = \frac{ydx - xdy}{x^2 + 2y^2}$? 若存在, 求出 $z(x, y)$, 若不存在, 说明理由.

$$1) P = \frac{y}{x^2 + 2y^2} \quad Q = \frac{-x}{x^2 + 2y^2}$$

$$\frac{\partial P}{\partial y} = \frac{x^2 + 2y^2 - y \cdot 4y}{(x^2 + 2y^2)^2} = \frac{x^2 - 2y^2}{(x^2 + 2y^2)^2}$$

$$\frac{\partial Q}{\partial x} = \frac{-1(x^2 + 2y^2) - (-x) \cdot 2x}{(x^2 + 2y^2)^2} = \frac{x^2 - 2y^2}{(x^2 + 2y^2)^2}$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \therefore \text{积分与路径 } L \text{ 的选取无关}$$

12) 存在

$$\frac{\partial z}{\partial x} = \frac{y}{x^2 + 2y^2}$$

$$z_1 = \int \frac{y}{x^2 + 2y^2} dx = \frac{\sqrt{2} \cdot y}{2y^2} \int \frac{1}{1 + (\frac{x}{\sqrt{2}y})^2} d\frac{x}{\sqrt{2}y}$$

$$= \frac{\sqrt{2}}{2} \int \frac{1}{1 + t^2} dt = \frac{\sqrt{2}}{2} \arctan\left(\frac{x}{\sqrt{2}y}\right)$$

$$z = z_1 + z(y)$$

$$\frac{-x}{x^2 + 2y^2} = \frac{\sqrt{2}}{2} \cdot \frac{\frac{x}{\sqrt{2}} \cdot (-\frac{1}{y^2})}{1 + \frac{x^2}{2y^2}} + z'(y)$$

$$\frac{-x}{x^2+y^2} = \frac{-x}{x^2+y^2} + z'(y) \quad z(y) = C$$

$$z(x,y) = \frac{\sqrt{2}}{2} \arctan \frac{x}{\sqrt{2}y} + C$$

4. 设 S^+ 为包含两点 $(-1, 0, 0), (1, 0, 0)$ 在内的光滑闭曲面, 正方向向外, 设向量值函数 $F(x, y, z) = \frac{(x, y, z) - (-1, 0, 0)}{\|(x, y, z) - (-1, 0, 0)\|^3} - 2 \frac{(x, y, z) - (1, 0, 0)}{\|(x, y, z) - (1, 0, 0)\|^3}$, 求 $\int_{S^+} F \cdot d\vec{S}$.

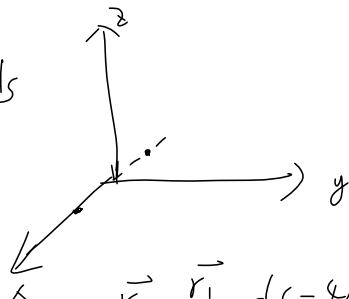
第二类曲面积分

$$\int_{S^+} F \cdot d\vec{S} = \int_{\partial B(p, \varepsilon)} \frac{\vec{r}_1}{|\vec{r}_1|^3} dS$$

$$- 2 \int_{\partial B(p_2, \varepsilon)} \frac{\vec{r}_2}{|\vec{r}_2|^3} dS$$

$$\int_{\partial B(p_1, \varepsilon)} \frac{\vec{r}_1}{|\vec{r}_1|^3} dS = \frac{1}{\varepsilon^3} \int_{\partial B(p_1, \varepsilon)} \vec{r}_1 \cdot \frac{\vec{r}_1}{\varepsilon} dS = 4\pi$$

$$\therefore \int_{S^+} F \cdot d\vec{S} = -4\pi$$



5. 设 $D = \{(x, y) : x^2 + y^2 \leq 1\}$, $f \in C^2(D)$, 且满足 $\Delta f = e^{-x^2-y^2}$, 求 $\int_D x f'_x + y f'_y dx dy$.

$$\begin{aligned} \int_D x f'_x + y f'_y dx dy &= \frac{1}{2} \int_D \nabla(x^2+y^2) \cdot \nabla f dx dy \\ &= -\frac{1}{2} \int_D (x^2+y^2) \Delta f dx dy + \frac{1}{2} \int_D \frac{\partial f}{\partial n} (x^2+y^2) dl \\ \int_D (x^2+y^2) \Delta f dx dy &= \int_0^{2\pi} \int_0^1 r^3 e^{-r^2} dr d\theta = \pi(1-e^{-1}) \\ \int_D \frac{\partial f}{\partial n} (x^2+y^2) dl &= \int_{\partial D} \frac{\partial f}{\partial n} dl = \int_{\partial D} \Delta f dx dy \\ &= \pi(1-e^{-1}) \\ \int_D x f'_x + y f'_y dx dy &= \frac{\pi}{2} \end{aligned}$$

6. 判断函数列 $f_n(x) = (1 + \frac{x}{n})^n$ 在 $[a, b]$ 和 $(-\infty, +\infty)$ 的一致收敛性.

$$\begin{aligned} \lim_{n \rightarrow \infty} f_n(x) &= e^x \quad \text{记 } f(x) = e^x \\ \forall x \in [a, b] \quad \forall n > \max\{-a, 1\}. \\ |f_n(x) - f(x)| &\leq e^b / e^{n \ln(1 + \frac{x}{n}) - x} - 1 \\ h(y) &= n \ln(1 + \frac{y}{n}) - y \quad h'(y) = \frac{-y/(y+n)}{(y+n)^2} \\ f_n(0) \nearrow (0, +\infty) \downarrow h(y) \\ n \geq a \text{ 时 } \max |1 - e^{h(y)}| &= \max \{1 - e^{h(a)}, 1 - e^{h(b)}\} \\ \text{代回: } |f_n - f| &\leq e^b \max \{1 - e^{h(a)}, 1 - e^{h(b)}\} \\ &= e^b \max \{1 - \frac{e^a}{(1 + \frac{a}{n})^n}, 1 - \frac{(1 + \frac{b}{n})^n}{e^n}\} \rightarrow 0 \\ \therefore f_n &\text{ 在 } [a, b] \text{ 上一致收敛于 } f. \\ \forall n > 0, \text{ 取 } x_n = n, \quad |f_n(x_n) - f(x_n)| &= (2^n - e^n) \geq e^{-2} \\ \therefore f_n &\text{ 在 } (-\infty, +\infty) \text{ 上不一致收敛} \end{aligned}$$

7. (1) 设幂级数 $\sum_{n=0}^{\infty} a_n x^n$ 和 $\sum_{n=0}^{\infty} b_n x^n$ 的收敛半径分别为 $R_1, R_2, R = \min\{R_1, R_2\}$, 求证在 $(-R, R)$ 中有

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} c_n x^n,$$

其中 $c_n = \sum_{k=0}^n a_k b_{n-k}$.

- (2) 已知级数 $\sum_{n=0}^{\infty} u_n$, $\sum_{n=0}^{\infty} v_n$ 以及他们的柯西乘积 $\sum_{n=0}^{\infty} w_n$ ($w_n = \sum_{k=0}^n u_k v_{n-k}$) 都收敛, 求证

$$\sum_{n=0}^{\infty} w_n = \left(\sum_{n=0}^{\infty} u_n \right) \left(\sum_{n=0}^{\infty} v_n \right).$$

$$\begin{aligned} (1) \quad & \left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{n=0}^{\infty} b_n x^n \right) = \left(\sum_{n=0}^{\infty} a_n x^n \right) \left(\sum_{k=0}^{\infty} b_k x^k \right) \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} a_n b_k x^{n+k} = \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} a_k b_{n-k} x^n \end{aligned}$$

(2) 上式令 $x=1$

8. 设 $f(x) = |x|$ ($-\pi \leq x \leq \pi$)

- (1) 求 f 在 $[-\pi, \pi]$ 上的 Fourier 级数.

(2) 计算 $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$.

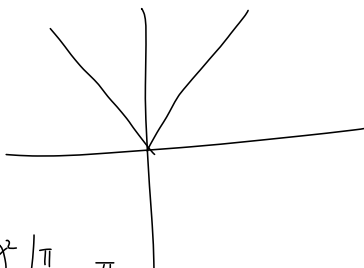
(3) 计算 $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4}$.

偶函数

$$\begin{aligned} (1) \quad a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{\pi} \cdot 2 \int_0^{\pi} x dx = \frac{2}{\pi} \cdot \frac{x^2}{2} \Big|_0^{\pi} = \pi \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx = \frac{2}{\pi n} \int_0^{\pi} x d(\sin nx) \\ &= \frac{2}{\pi n} \cdot \left[x \sin nx \Big|_0^{\pi} - \int_0^{\pi} \sin nx dx \right] \\ &= \frac{-2}{\pi n} \cdot \int_0^{\pi} \sin nx dx = \frac{-2}{\pi n} \cdot \frac{\cos nx}{-n} \Big|_0^{\pi} = \frac{2}{n^2 \pi} (\cos n\pi - 1) \end{aligned}$$

n 为奇 -2 n 偶 0



$$= \frac{2}{n^2 \pi} \left[(-1)^n - 1 \right]$$

$$f(x) \sim \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1] \cos nx}{n^2 \pi}$$

$$(2) f(x) \sim \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2 \cdot (-2) \cos nx}{(2n-1)^2 \pi}$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{-4 \cos nx}{(2n-1)^2 \pi}$$

$$x=0$$

$$-\frac{\pi}{2} = \sum_{n=1}^{\infty} \frac{-4}{(2n-1)^2 \pi}$$

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$8 (1) f(x) \sim \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}$$

$$(2) \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

$$(3) \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$